

Business Memo: Instacart Order Analysis

Product Growth Opportunities

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Executive Summary

This analysis was conducted on 3.4 million grocery orders to identify growth levers for the product. The key findings reveal that weekend shoppers generate significantly higher order values, staple products drive customer loyalty, and a meaningful segment of users is at risk of churning. Immediate action on these three fronts can drive measurable revenue growth.

Key Insights & Recommendations

1. Weekend Conversion Lift (A/B Test Simulation)

Finding: Weekend shoppers have an average basket size of **11.46 items**, compared to **10.06** on weekdays. This is a **14% increase**.

Statistical Validation: An independent T-test yielded a P-value of **0.0000000000**, confirming this difference is statistically significant and not due to random chance.

Recommendation: Invest in weekend-specific marketing campaigns, push notifications, and UI features (e.g., “Weekend Family Bundles”) to capitalize on this behavior. The 14% lift translates to significant incremental revenue.

2. Retention Drivers (Best Sellers)

Finding: Staple grocery items command the highest reorder rates.

- **Bananas:** 18,726 purchases, 88.4% reorder rate.
- **Organic Low Fat Milk:** 91.3% reorder rate.

Recommendation: Guarantee 100% stock availability for these staples. Out-of-stock events on these items directly threaten customer loyalty. These products should be prioritized in warehouse restocking schedules.

3. Customer Health (RFM Segmentation)

Finding: Currently:

- **53,622 Champions** (high-value, recent buyers).
- **19,950 At-Risk** users (low engagement, likely to churn).

Recommendation: Launch a targeted “Win-Back” campaign specifically for the At-Risk segment. Offering a 10% discount voucher or free delivery could reactivate these users, increasing overall revenue without the cost of acquiring new customers.

4. Overall KPIs

The following key performance indicators provide a baseline for the business health.

Next Steps

1. **Feature Development:** Run a live randomized A/B test for the proposed “Weekend Bundle” feature to confirm causality.
2. **Marketing:** Deploy the win-back campaign within the next 14 days to reduce churn.

Metric	Value
Total Orders	131,209
Unique Customers	131,209
Avg Basket Size	10.55 items
Avg Days Between Orders	17.07 days

3. **Dashboard:** A live dashboard tracking these KPIs is available [at this link](#).

Appendix

Tools Used: PostgreSQL, Python (Pandas, Scipy, Matplotlib), Looker Studio.

Data Sample: Instacart Market Basket Dataset (3.4M orders, 130k users).

Repository: [GitHub Repository Link](#)